

Package ‘scScale’

July 8, 2026

Title Single-Cell Scaling Law Utilities

Version 0.0.1

Author Jingyuan Hu [aut, cre],
Sijia Zhu [aut, cre]

Maintainer Jingyuan Hu <jingyuanhu@example.com>

Description Median Marchenko-Pastur noise calibration, BBP spike inversion,
spectral overlap, and task-difficulty utilities for single-cell scaling
analyses, aligned with ChesleaZ/scScale.

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URL <https://github.com/ChesleaZ/scScale>

BugReports <https://github.com/ChesleaZ/scScale/issues>

Encoding UTF-8

Depends R (>= 2.10),
Matrix

LazyData true

LazyDataCompression xz

Suggests irlba,
SeuratObject

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

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gse123025_myeloid_hvg *GSE123025 Myeloid HVG Count Matrix Example*

Description

A larger example subset derived from GSE123025 for RMT spike-model tutorials. The object keeps all cells from the source matrix and restricts genes to 2,000 highly variable genes selected by `select_hvgs()`.

Usage

```
data(gse123025_myeloid_hvg)
```

Format

A list with:

counts Integer matrix with 2,000 selected genes/features by 1,922 cells.

source Source file name in the canonical lab data directory.

accession GEO accession identifier, "GSE123025".

description Short description of the HVG subset.

selection List describing the feature and cell selection.

original_dim Dimensions of the source matrix before HVG selection.

cell_totals Column sums of the HVG count matrix.

gene_totals Row sums of the HVG count matrix.

Source

Derived from `data/GSE123025/GSE123025_Single_myeloid_1922_cells_processed_data.csv.gz` in the canonical lab workspace. The source matrix has 23,429 genes by 1,922 cells and is about 8.0 MB compressed as CSV gzip, or 95 MB uncompressed.

Examples

```
data(gse123025_myeloid_hvg)
counts <- gse123025_myeloid_hvg$counts
dim(counts)
fit <- scscale_fit(counts, r = 10, mp_max_iter = 3, fit_umi = FALSE)
fit
```

`gse164378_3p_citeseq_hvg`*GSE164378 3 Prime CITE-seq RNA HVG and ADT Example*

Description

A compact paired RNA and ADT example derived from the 3 prime CITE-seq portion of GSE164378. The object keeps 4,000 matched PBMC cells, 2,000 RNA highly variable genes, and all 228 measured ADT features.

Usage

```
data(gse164378_3p_citeseq_hvg)
```

Format

A list with:

rna_counts Sparse integer matrix with 2,000 selected RNA genes by 4,000 cells.

adt_counts Sparse integer matrix with 228 ADT features by the same 4,000 cells.

meta Cell metadata, including RNA/ADT QC totals, lane, donor, time, batch, and cell type labels.

source Short source dataset description.

accession GEO accession identifier, "GSE164378".

description Short description of the compact paired-modality subset.

selection List describing the cell and RNA feature selection.

original_dim Dimensions of the source 3 prime RNA and ADT matrices before subsetting.

cell_totals RNA and ADT column sums for the selected cells.

feature_totals RNA and ADT row sums for the selected features.

Source

Derived from the 3 prime RNA and ADT matrices in GSE164378, the Hao et al. multimodal PBMC reference dataset. The source 3 prime matrices have 33,538 RNA features by 161,764 cells and 228 ADT features by the same 161,764 cells.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts
adt <- gse164378_3p_citeseq_hvg$adt_counts
dim(rna)
dim(adt)
```

scScale

*Single-Cell Scaling Law Utilities***Description**

Gaussian spike fitting and low-rank spectral mutual information utilities for single-cell scaling-law analyses.

Details

The recommended paired-modality workflow is:

1. Fit matched modalities once with [scscale_pair_fit](#).
2. Fit UMI-depth scaling with [scscale_umi_mi](#).
3. Fit cell-number scaling with [scscale_cell_number_mi](#).
4. Fit the joint cell-number by UMI grid with [scscale_cell_number_by_umi_mi](#).
5. Compare with direct empirical MI from [scscale_empirical_mi](#).

All theory curves are routed through [scscale_low_rank_mi](#), the single low-rank subspace-alignment MI formula.

See Also

[scscale_fit](#), [scscale_pair_fit](#), [scscale_low_rank_mi](#), [scscale_umi_mi](#), [scscale_cell_number_mi](#), [scscale_cell_number_by_umi_mi](#), [scscale_empirical_mi](#)

scscale_cell_number_by_umi_mi

*Fit Joint Cell-Number by UMI MI Scaling***Description**

Compute the theory MI grid over both cell number and UMI depth, reusing the refitted UMI spike strengths and subspace alignments.

Usage

```
scscale_cell_number_by_umi_mi(pair, umi, n_grid,
  sampling_rates = umi$sampling_rates)
```

Arguments

`pair` A [scscale_pair_fit](#) object.
`umi` A [scscale_umi_mi](#) object.
`n_grid` Cell-number grid.
`sampling_rates` Subset or ordering of UMI sampling rates to use.

Value

A data frame with `n`, `c_X`, `I_theory`, and `sampling_rate`.

scscale_cell_number_mi

Fit Cell-Number MI Scaling

Description

Compute the theory MI curve as the number of cells varies.

Usage

```
scscale_cell_number_mi(pair, n_grid,
  q_X = pair$x_fit$spikes$q_X[seq_len(pair$r_X)], P = pair$P)
```

Arguments

pair	A <code>scscale_pair_fit</code> object.
n_grid	Cell-number grid.
q_X	Optional X-side spike-strength vector.
P	Optional X/Y low-rank subspace alignment matrix.

Value

A data frame with n, c_X, and I_theory.

scscale_empirical_inu_grid

Sample an Empirical I(n, U) Grid

Description

Sample cells and sequencing depth, then compute fixed-rank empirical subspace MI for each (n, U) grid point. This is the direct empirical counterpart to spike-model scaling workflows.

Usage

```
scscale_empirical_inu_grid(x, target, n_grid, U_grid = NULL,
  sampling_rates = NULL, r = 10, n_replicates = 1, seed = 1,
  input = c("counts", "normalized"), target_input = input,
  target_depth = 1e4, count_transform = c("log1p_cpm", "pearson_residual",
  "log1p"), center = TRUE, scale = FALSE, eps = 1e-12,
  use_irlba = TRUE)
```

Arguments

x	Feature-by-cell source matrix.
target	Feature-by-cell target matrix or cell-level target vector.
n_grid	Cell numbers to sample.
U_grid	Target UMI depths per cell. Provide either U_grid or sampling_rates.
sampling_rates	Count downsampling fractions in (0,1].
r	Fixed rank used for empirical subspace MI.
n_replicates	Number of replicate subsamples per grid point.
seed	Random seed base.
input, target_input	Input type for x and matrix targets.
target_depth, count_transform, center, scale	Preprocessing controls passed to empirical MI calculation.
eps	Numerical clipping value for MI.
use_irlba	Whether to use irlba for partial SVD when available.

Value

A data frame with sampled n, requested U, replicate index, I_empirical, effective MI rank, and observed UMI summaries.

scscale_empirical_mi *Compute Empirical Subspace Mutual Information*

Description

Compute empirical low-rank subspace MI directly from data, either between two matrices or between one matrix and a cell-level target.

Usage

```
scscale_empirical_mi(x, target, r = 10, input = c("counts", "normalized"),
  target_input = input, target_depth = 1e4, count_transform = c("log1p_cpm",
    "pearson_residual", "log1p"), center = TRUE, scale = FALSE,
  eps = 1e-12, use_irlba = TRUE, store_subspaces = FALSE)
```

Arguments

x	Feature-by-cell matrix for the source modality.
target	Second feature-by-cell matrix or cell-level target vector.
r	Number of singular-vector directions to keep.
input, target_input	Input type for x and matrix targets.
target_depth, count_transform, center, scale	Preprocessing controls.
eps	Numerical clipping value.
use_irlba	Whether to use irlba for partial SVD when available.
store_subspaces	Whether to keep fitted subspace matrices in the output.

Value

A list containing empirical mi, canonical-correlation-style gamma, and effective rank information.

scscale_empirical_scaling_fit

Fit Direct Empirical Scaling Laws

Description

Fit direct nonlinear scaling laws to fixed-rank empirical subspace MI values, without spike calling or Marchenko–Pastur bulk fitting. The cell law is

$$I(n) = I_{\infty|U_0} - C_n \log(1 + A_n/n).$$

The UMI law is

$$I(U) = I_{\infty|n_0} - C_U \log(1 + A_U/(U + U_0)).$$

The joint law is additive in the two empirical information gaps:

$$I(n, U) = I_{\infty} - C_n \log(1 + A_n/n) - C_U \log(1 + A_U/(U + U_0)).$$

Usage

```
scscale_fit_empirical_cell_law(data, n_col = "n", mi_col = "I_empirical",
  start = NULL, control = list(maxit = 10000))

scscale_fit_empirical_umi_law(data, U_col = "U", mi_col = "I_empirical",
  start = NULL, control = list(maxit = 10000))

scscale_fit_empirical_joint_law(data, n_col = "n", U_col = "U",
  mi_col = "I_empirical", cell_fit = NULL, umi_fit = NULL,
  mode = c("hybrid", "transfer", "free"), start = NULL,
  control = list(maxit = 20000))

scscale_empirical_global_ceiling(cell_fit, umi_fit, n0, U0)

scscale_empirical_cell_curve(n, I_infinity, C_n, A_n)

scscale_empirical_umi_curve(U, I_infinity, C_U, A_U, U_offset)

scscale_empirical_joint_curve(n, U, I_infinity, C_n, A_n, C_U, A_U,
  U_offset)
```

Arguments

data	Data frame containing empirical MI observations.
n_col, U_col, mi_col	Column names for cell number, UMI depth, and empirical MI.
start	Optional named list of starting values.
control	Control list passed to optim .

cell_fit, umi_fit	Marginal empirical scaling fits used to initialize or transfer joint-law parameters.
mode	Joint fitting mode. "transfer" fixes marginal gap parameters and fits only the global ceiling; "hybrid" fixes scale parameters and refits the ceiling and amplitudes; "free" refits all joint parameters.
n, U	Cell number and UMI depth values for prediction.
n0, U0	Reference cell number and UMI depth for extrapolating the global ceiling from marginal fits.
I_infinity, C_n, A_n, C_U, A_U, U_offset	Scaling-law parameters.

Value

The fitting functions return an object of class `scscale_empirical_scaling_fit`, with fitted coefficients, residuals, fit metrics, and `predict()` support. `scscale_empirical_global_ceiling` returns the two marginally implied global ceilings and their average.

<code>scscale_fit</code>	<i>Fit a Gaussian Spike Model</i>
--------------------------	-----------------------------------

Description

Fit the input matrix with the Gaussian spike model, estimate MP bulk and spike parameters, and optionally attach target and UMI-scaling summaries.

Usage

```
scscale_fit(x, input = c("counts", "normalized", "representation",
  "eigenvalues"), target = NULL, target_fit = NULL, n = NULL, p = NULL,
  target_depth = 1e4, count_transform = c("log1p_cpm",
  "pearson_residual", "log1p"), center = TRUE, scale = FALSE,
  n_features = NULL, min_cells = 10, mp_max_iter = 300, mp_grid_n = 20000,
  mp_stop_metric = c("qq_rmse_log", "ks", "neg_log_likelihood", "none"),
  mp_stop_tol = 1e-4, mp_min_iter = 2, mp_gap_z_threshold = 6,
  mp_quiet_z_threshold = 5, mp_quiet_run = 20,
  mp_background_skip = 25, mp_background_window = 500,
  mp_central_quantiles = c(0.10, 0.70), r = NULL,
  force_spikes = FALSE,
  fit_umi = input[1] == "counts",
  sampling_rates = c(0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00),
  U_grid = NULL, umi_linear_intercept = NULL, umi_linear_slope = NULL,
  umi_reference_U = NULL, umi_clamp_rate = TRUE, umi_seed = 1,
  umi_replicates = 1, store_matrix = FALSE, use_irlba = TRUE)
```

Arguments

x	Feature-by-cell counts, normalized matrix, representation matrix, or eigenvalue vector.
input	Input type.

target, target_fit
Optional target inputs used by legacy MI summaries.

n, p
Cell and feature counts, required when input = "eigenvalues".

target_depth, count_transform, center, scale, n_features, min_cells
Count preprocessing controls.

mp_max_iter, mp_grid_n, mp_stop_metric, mp_stop_tol, mp_min_iter,
mp_gap_z_threshold, mp_quiet_z_threshold, mp_quiet_run,
mp_background_skip, mp_background_window, mp_central_quantiles
MAD/MP bulk and spike-fitting controls.

r
Optional number of spikes to keep.

force_spikes
Whether to force the retained top components to be treated as spikes.

fit_umi, sampling_rates, U_grid, umi_seed, umi_replicates
Legacy UMI scaling controls.

umi_linear_intercept, umi_linear_slope, umi_reference_U,
umi_clamp_rate
Optional controls for linearized UMI-rate calibration.

store_matrix
Whether to store the normalized matrix in the returned fit.

use_irlba
Whether to use **irlba** for partial SVD when available.

Value

An `scscale_fit` object containing the spectrum, MP bulk fit, spike table, recoverability vectors, and optional stored normalized matrix.

Examples

```
data(gse123025_myeloid_hvg)
x <- gse123025_myeloid_hvg$counts[1:100, 1:200]
fit <- scscale_fit(x, mp_grid_n = 600, mp_max_iter = 50)
head(fit$spikes)
```

scscale_low_rank_mi *Low-Rank Subspace Mutual Information*

Description

Compute the theory MI from X/Y recoverability and optional low-rank subspace alignment. This is the single theory formula used by the scaling helpers.

Usage

```
scscale_low_rank_mi(theta_X, theta_Y, P = NULL, eps = 1e-12)
```

Arguments

theta_X, theta_Y
Recoverability vectors for the X and Y low-rank signal directions.

P
Optional subspace alignment matrix between X and Y. If omitted, directions are matched by rank.

eps
Numerical clipping value.

Value

A list with `mi`, `I_theory`, singular values `sigma`, squared canonical correlations `gamma`, and effective ranks.

Examples

```
scscale_low_rank_mi(c(0.5, 0.2), c(0.4, 0.1))$mi
```

scscale_mi	<i>Compare Two Fitted scScale Objects</i>
------------	---

Description

Compare two fitted objects with `scscale_low_rank_mi` and optional legacy scaling summaries.

Usage

```
scscale_mi(fit, target_fit, n_grid = NULL, U_grid = NULL,
  sampling_rates = NULL, combine = TRUE, empirical = TRUE,
  store_empirical_subspaces = FALSE, use_irlba = TRUE, r = NULL,
  P = NULL, eps = 1e-12)
```

Arguments

<code>fit, target_fit</code>	<code>scscale_fit</code> objects.
<code>n_grid, U_grid, sampling_rates</code>	Optional cell-number and UMI grids.
<code>combine</code>	Whether to return a combined legacy grid when both dimensions are supplied.
<code>empirical</code>	Whether to compute empirical subspace MI from stored matrices.
<code>store_empirical_subspaces</code>	Whether to keep empirical subspace matrices.
<code>use_irlba</code>	Whether to use irlba for partial SVD when available.
<code>r</code>	Optional number of directions to keep.
<code>P</code>	Optional low-rank subspace alignment matrix.
<code>eps</code>	Numerical clipping value.

Value

A list containing theory MI and optional empirical or scaling components.

scscale_pair_fit

*Fit Matched Modalities for MI Scaling***Description**

Fit the noisy input modality, optionally fit or directly construct the target side, compute right singular vectors, and align the low-rank subspaces.

Usage

```
scscale_pair_fit(x, y, count_transform = c("log1p_cpm",
    "pearson_residual", "log1p"), mp_max_iter = 300, mp_grid_n = 3000,
    y_mode = c("spike", "subspace", "auto"), y_rank = NULL,
    y_center = TRUE, y_scale = FALSE, y_input = c("auto", "matrix",
    "labels", "numeric"), auto_low_rank_max = 50, rank_tol = 1e-08,
    x_rank = NULL, x_force_spikes = FALSE, use_irlba = TRUE, ...)
```

Arguments

x	Matched feature-by-cell count matrix for the noisy input modality.
y	Matched target. Usually a feature-by-cell count matrix for a second noisy modality, but can also be a deterministic low-rank target matrix or label/numeric vector when y_mode = "subspace".
count_transform	Count normalization transform passed to scscale_fit .
mp_max_iter, mp_grid_n	MAD/MP spike-fitting controls.
y_mode	How to handle y. "spike" preserves the original paired-modality behavior and fits a spike model to y. "subspace" treats y as a deterministic target subspace with target recoverability equal to one. "auto" uses conservative low-rank/one-hot detection.
y_rank	Optional target rank when y_mode = "subspace". Defaults to the numerical rank after preprocessing.
y_center, y_scale	Whether to center or scale deterministic target rows before taking right singular vectors.
y_input	Target type for deterministic targets. "auto" infers matrix, label, or numeric-vector input.
auto_low_rank_max, rank_tol	Controls for conservative automatic deterministic-target detection.
x_rank	Optional number of input spikes to retain.
x_force_spikes	Whether to force the retained input components to be treated as spikes.
use_irlba	Whether to use irlba for partial SVD when available.
...	Additional arguments passed to scscale_fit .

Value

A `scscale_pair_fit` object with `x_fit`, `y_fit`, `z_X`, `z_Y`, the overlap matrix `P`, baseline `mi`, and `I_infinity`. When `y_mode = "subspace"`, `y_fit` is a `scscale_subspace_fit` object and no MP spike model is fitted to the target.

Examples

```

data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[1:100, 1:250]
adt <- gse164378_3p_citeseq_hvg$adt_counts[1:35, 1:250]
pair <- scscale_pair_fit(rna, adt, mp_grid_n = 600, mp_max_iter = 50)
pair$I_infinity

labels <- factor(rep(c("A", "B"), each = 125))
label_pair <- scscale_pair_fit(
  rna,
  labels,
  y_mode = "subspace",
  y_input = "labels",
  mp_grid_n = 600,
  mp_max_iter = 50
)
label_pair$y_mode

```

```
scscale_subspace_overlap_matrix
```

Compute a Subspace Overlap Matrix

Description

Compute the cross-product matrix between two low-rank cell subspaces after optional cell-name alignment. This matrix is used by [scscale_low_rank_mi](#).

Usage

```
scscale_subspace_overlap_matrix(z_X, z_Y)
```

Arguments

<code>z_X, z_Y</code>	Cell-by-rank orthonormal subspace bases. If row names are present, bases are aligned by shared cell names.
-----------------------	--

Value

A numeric overlap matrix with rows corresponding to `z_X` directions and columns corresponding to `z_Y` directions.

```
scscale_target_fit
```

Construct a Deterministic Target Subspace Fit

Description

Construct a deterministic low-rank target object for use in [scscale_pair_fit](#). Unlike [scscale_fit](#), this function does not fit an MP/BBP spike model. It computes the target right singular vectors directly and sets target recoverability to one.

Usage

```
scscale_target_fit(y, input = c("matrix", "labels", "numeric"),
  rank = NULL, center = TRUE, scale = FALSE, drop_empty = TRUE,
  rank_tol = 1e-08, use_irlba = TRUE)
```

Arguments

y	Target matrix, label vector, or numeric vector. Matrix inputs are feature-by-cell.
input	Target input type.
rank	Optional number of target singular vectors. Defaults to the numerical rank after preprocessing.
center, scale	Whether to center or scale target rows before computing the target subspace. Label targets are typically centered and not scaled.
drop_empty	Whether to drop unused label levels.
rank_tol	Numerical rank tolerance.
use_irlba	Whether to use irlba for partial SVD when available.

Value

A `scscale_subspace_fit` object with target matrix `matrix`, cell subspace `z`, rank `r`, and unit recoverability vectors `theta_X` and `theta_infinity`.

Examples

```
labels <- factor(rep(c("T", "B", "Mono"), length.out = 90))
target <- scscale_target_fit(labels, input = "labels")
target$r
```

scscale_umi_mi	<i>Fit UMI-Depth MI Scaling</i>
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Description

Refit the X modality over UMI depth, recompute the X/Y subspace alignment, and return both the refitted theory MI curve and the refitted `I_infinity` bound.

Usage

```
scscale_umi_mi(pair, x_counts, sampling_rates = c(0.10, 0.20, 0.35,
  0.50, 0.70, 0.85, 1.00),
  count_transform = pair$count_transform %||% "log1p_cpm", seed = 1,
  mp_max_iter = 300, mp_grid_n = 2000, use_irlba = TRUE)
```

Arguments

`pair` A `scscale_pair_fit` object.
`x_counts` Original X-side count matrix to downsample and refit.
`sampling_rates` Global UMI sampling fractions in $(0, 1]$.
`count_transform` Count transform used for refits.
`seed` Random seed for downsampling.
`mp_max_iter, mp_grid_n` MAD/MP spike-fitting controls for refits.
`use_irlba` Whether to use **irlba** for partial SVD when available.

Value

A `scscale_umi_mi` object with `curve`, `q_by_rate`, `fit_by_rate`, and `P_by_rate`.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[1:100, 1:250]
adt <- gse164378_3p_citeseq_hvg$adt_counts[1:35, 1:250]
pair <- scscale_pair_fit(rna, adt, mp_grid_n = 600, mp_max_iter = 50)
umi <- scscale_umi_mi(pair, rna, sampling_rates = c(0.5, 1),
  mp_grid_n = 500, mp_max_iter = 50)
umi$curve
```

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